Algorithm Analysis

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Algorithm

Problem

Solution

Which algorithm to solve?

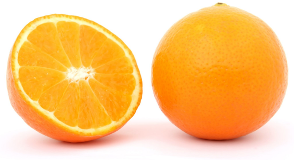
Algorithm

Which algorithm is the best?

How do we know that the algorithm is good?

Computational Complexity

- If computers were infinitely fast, **any correct method** for solving a problem would do.
- Computing time and space in memory are a limited resource.
- Algorithms that are efficient in terms of **time or space** are preferred.



→ Algorithm →

Computational Complexity

- How do we measure algorithm computational efficiency?
- How do we measure algorithm’s resources usage?
- How do we measure algorithm’s **efficiency** or **performance**?

Computational Complexity

1. Use running time as an indicator.

Example - Summation of *n* integers

```
def sumOfN(n:int):
    total = 0
    for i in range(1, n+1):
        total = total + i
    return total
```

Running Time of Summation

- For loop version

	Time
sumOfN(10000)	
sumOfN(100000)	
sumOfN(1000000)	

Which run should be slowest?

Example - Summation of *n* integers

```python
def sumOfN(n:int):
    total = 0
    for i in range(1, n+1):
        total = total + i
    return total
```

- Time required seems to increase as we increase the input size (*n*).

Running Time of Summation

- For loop version

	Time
sumOfN(10000)	
sumOfN(100000)	
sumOfN(1000000)	

- Formula based version

	Time
sumOfN2(10000)	
sumOfN2(100000)	
sumOfN2(1000000)	

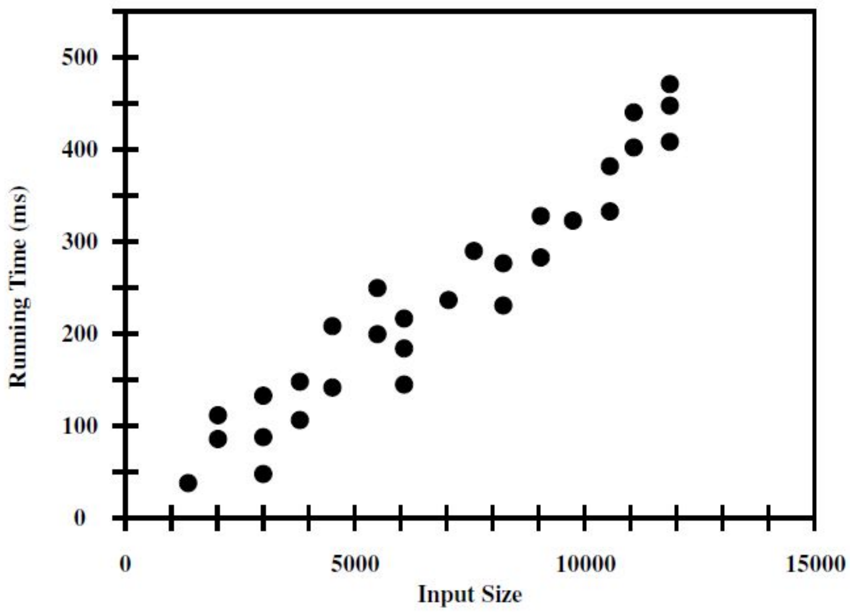
Is formula based version faster than for loop version?

Running Time

- Input size
- Running time also depends on many factors
    - Hardware (CPU, RAM, etc.)
    - Software (OS, Programming language, etc.)

Experimental Studies

- Implement an algorithm then study its running time by
  - Executing with different test inputs of various sizes and
  - Recording time spent for each input size
  - Plot the results



Challenges of Experiments

- To directly compare between two different algorithms, the same hardware and software environments must be used.
- Limited set of test inputs.
- An algorithm must be fully implemented to study its running time.
  - Can be costly for some algorithms

High-level Analysis

- Instead of implementing an algorithm and perform experiments, we can study a high-level description of the algorithm.
  - Either in the form of an actual code or pseudo-code.
- Takes into account all possible inputs
- Allows us to evaluate the efficiency of an algorithm independent of hardware & software environment.

Computational Complexity

1. Use running time as an indicator.
2. Use number of operations or steps as an indicator.

Counting Primitive Operations

- Formally, a primitive operation corresponds to a low-level instruction with an execution time that is constant.
    - Assign a variable to an object
    - Determining the object associated with a variable
    - Performing an arithmetic operation
    - Comparing two numbers
    - Accessing a single element of a Python list by an index
    - Calling a function (excluding operations within the function)
    - Returning from a function

Counting Operations

```
def sumOfN(n:int):
    total = 0
    for i in range(1, n+1):
        total = total + i
    return total
```

Number of Operations

Time complexity

$T(n) =$

Seven Important Functions

- Seven fundamental functions in algorithm analysis:
  - Constant: $f(n) = c$
  - Logarithmic: $f(n) = \log_b n$
  - Linear: $f(n) = n$
  - N-Log-n: $f(n) = n \log n$
  - Quadratic: $f(n) = n^2$
  - Cubic: $f(n) = n^3$
  - Exponential: $f(n) = 2^n$

Time Complexity

- To see long-term / big picture trends of running time
    - Given an algorithm that takes *input size* ***n***, find a function **T(*n*)** that describes the *runtime* of the algorithm

Loops

	Python Code	Time complexity
Linear	for i in range (0, n): do something	$T(n) =$
Linear	for i in range (0, n, 2): do something	$T(n) =$

Nested	for i in range (0, n):
	for j in range (0, n):
	do something
		$T(n) =$

A log is a Repeated Division

Python Code

Time complexity

n=1000

Logarithmic

```
i = 1              Start
while i < n:       Stop
    do something
    i = i * 2      Step
```

1
2
4
8
16
32
64
128
256
512

$T(n) =$

Logarithmic

```
i = n              Stop
while i >= 1:      Start
    do something
    i = i // 2     Step
```

1000
500
250
125
62
31
15
7
3
1

$T(n) =$

Linear Logarithmic Loops

Python Code

Time complexity

Linear logarithmic

```
for i in range (0,n):       n
    j = 1
    while j < n:            log n
        do something
        j = j*2
```

$T(n) =$

Quadratic Loops

Python Code

Time complexity

Dependent Nested

```
for i in range (0,n):  n
    for j in range(0, i+1): (n+1)/2
        do something
```

$T(n) =$

Number of iterations of the inner loop *j* depends on the outer loop *i*

For the inner loop *j*, the number of iterations for each *i* on average is (n+1)/2

For example, n = 3, i = 0 then j = [0],
i = 1 then j = [0, 1],
i = 2 then j = [0, 1, 2]

Exponential

Python Code

Time complexity

Double Recursive

```
def foo(n):
    if (n==1):
        Return True
    foo(n-1)
    foo(n-1)
```

$T(n) =$

Input Size Considerations

- *Input size* might be:
  - the *magnitude of the input value* (e.g., for numeric input)
  - the *number of items* in the input (e.g., as in a list)
- An algorithm may also be dependent on *more than one input*.

Computational Complexity

1. Use running time as an indicator.
2. Use number of operations or steps as an indicator.
    - Represented by T(*n*)

Do we need to count a precise number of operations?

Asymptotic Notation

$T(n) = 2n^2 + n + 1$

- The running time of this algorithm grows as $\boldsymbol{n^2}$.
- $T(n) \approx 2\boldsymbol{n^2} \approx \boldsymbol{n^2}$
- T(n) is asymptotic to $\boldsymbol{n^2}$
  - When n approaches to infinity, the rests are **insignificant**.

Asymptotic Notation

- Asymptotic notation represents algorithm’s complexity.
  - Ignores constant factors and slower growing terms.
  - Focus on the main components that affect the growth.
  - Big-O notation

Computational Complexity

1. Use running time as an indicator.
2. Use number of operations or steps as an indicator.
    - Represented by $T(n)$
3. Use asymptotic notation as an indicator.
    - Big-O notation

Big-O Notation

- Objectively describe the efficiency of code without the use of concrete units (seconds/bytes).
- Provide a big picture of how the time and space requirements scale w.r.t input size.
- Focus on worst-case scenario.

Big-O Notation

- Formally, f(n) = O(g(n)):
    - If there exist positive constants c and n_0
    - such that $0 < f(n) < c * g(n)$ for all $n >= n_0$
- f(n) is a big-O of g(n)
    - Intuitively means that g (multiplied by a constant factor) set an *upper bound* of f as n gets large - i.e., an *asymptotic bound*

Big-O Notation

Example: $f(n) = 4n^3 + 3n^2 + 5$

- Need to find $g(n)$, c, n_0 that satisfies $0 < f(n) < c * g(n)$ for all $n >= n_0$

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Simplifying Big-O

- Product Rule
  - If the Big-O is the product of the multiple terms, **drop the constant terms**

$O(1024 * n) =$

$O(n / 10) =$

$O(7 * n * n) =$

$O(345) =$

Simplifying Big-O

- Sum Rule
  - If the Big-O is the sum of the multiple terms, **only keep the largest term, drop the rest.**

$O(100 + n) =$

$O(n^2 + n) =$

$O(n + 500 + n^3 + n^2) =$

Big-O Notation

- Example:
  - $T(n) = 1 + n$,

    then $T(n) =$
  - $T(n) = 5n^2 + 10n + 12$,

    then $T(n) =$

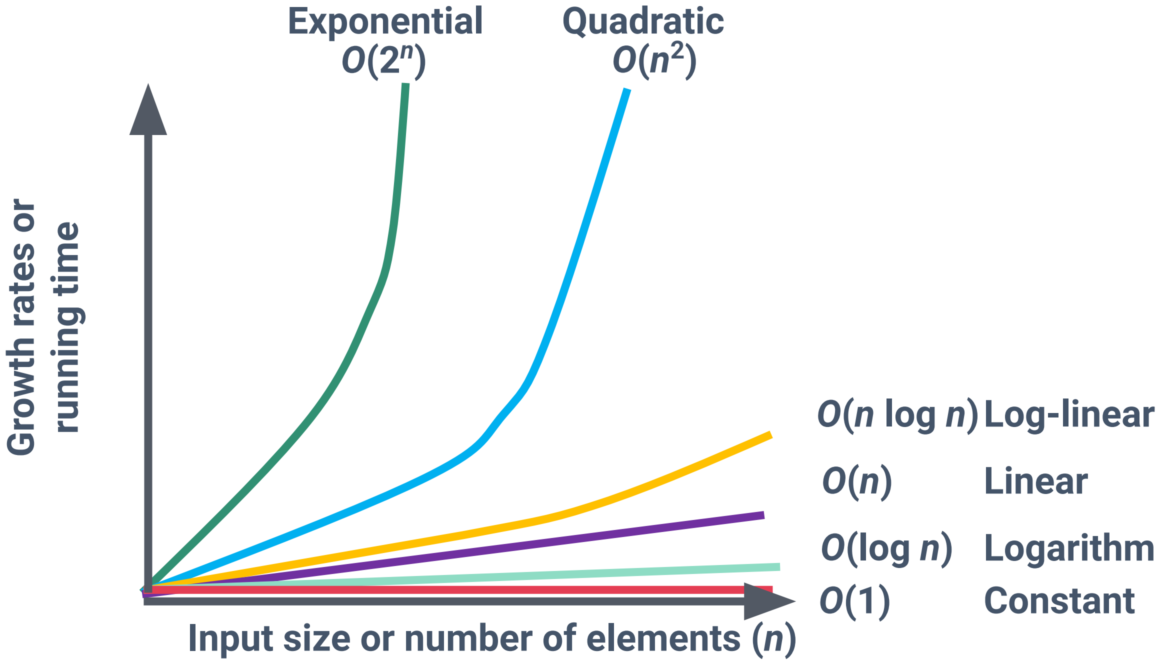
- **$O(n^2)$ means time complexity $T(n)$ will never exceed $O(n^2)$.**

Big-O Notation (Ordered)

- Common f(*n*):
  - O(1): constant
  - O(log *n*): logarithm
  - O(*n*): Linear
  - O(*n* log *n*): Log linear
  - O(n^2): Quadratic
  - O(2^n): Exponential
  - O(*n*!): Factorial

Time Complexity

In general, the standard functions of input size n are shown in figure.



Big-O Exercise

Time complexity T(*n*)

- $n^2 + 100n$
- $n^2*n + 100n^2\log n$
- $123 + \log 657$
- $(n + \log n)^3$
- $n(2 + \log n)$
- $(n/3)^6 + 10n$
- $1 + 2 + 3 + ... + n$

Big-O

Exercise: Calculating Time Complexity

Python Code: Factorial

```python
def factorial1(n):
  if n <= 1:
    return 1
  else:
    fact = 1
    for i in range(2,n+1):
      fact *= i
    return fact
```

Exercise: Calculating Time Complexity

Python Code: Simple nested loops

```
def simple(n):
  for i in range(n):
    for j in range(n):
      print("i: {0}, j: {1}".format(i,j))
```

Exercise: Calculating Time Complexity

Python Code: Element uniqueness v1

```python
def unique1(s):
  for i in range(len(s)):
    for j in range(i+1, len(s)):
      if s[i] == s[j]:
        return False  # Found duplicate pair
  return True       # All elements are unique
```

Exercise: Calculating Time Complexity

Python Code: Prefix averages v1

```python
def prefix_average1(s):
  n = len(s)
  a = [0] * n                     # create list of n zeros
  for i in range(n):
    total = 0                     # compute each element
    for j in range(i+1):
      total += s[j]
    a[i] = total / (i+1)          # record the average
  return a
```

Exercise: Calculating Time Complexity

Python Code: Prefix averages v2

```
def prefix_average2(s):
  n = len(s)
  a = [0] * n                  # create list of n zeros
  total = 0
  for i in range(n):
    total += s[i]     # update total sum to include s[i]
    a[i] = total / (i+1) # compute average based on
current sum
  return a
```

Algorithm Computational Complexity

1. Use running time as an indicator.
2. Use number of operations or steps as an indicator.
   - Represented by T(*n*)
3. Use asymptotic notation as an indicator.
   - Big-O notation